

Denote by $p(n)$ the number of unrestricted partitions of n , that is

$$f(x) = 1 + \sum_{n=1}^{\infty} p(n)x^n = \frac{1}{(1-x)(1-x^2)(1-x^3)\dots}$$

Prove that as $n \rightarrow \infty$,

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

Ramanujan's Circle Method

Warm up: $\log p(n) \sim C\sqrt{n}$

Elementary calculus shows that $\log p(n) \sim C\sqrt{n}$ where $C = \pi\sqrt{2/3}$.

Recall a Tauberian theorem:

Theorem: If $g(x) = \sum a_n x^n$ is a series where $a_n \geq 0$, and $\log g(x) \sim \frac{A}{1-x}$ when $x \rightarrow 1^-$, then $\log s_n = \log(a_0 + \dots + a_n) \sim 2\sqrt{An}$.

Now consider $g(x) = (1-x)f(x) = \sum (p_n - p_{n-1})x^n$, then $g(x) = \frac{1}{(1-x^2)(1-x^3)\dots}$ so

$$\log g(x) \sim \sum_{k=2}^{\infty} -\log(1-x^k) = \sum_{k=2}^{\infty} \sum_{j=1}^{\infty} \frac{x^{kj}}{j} = \sum_{j=1}^{\infty} \frac{x^{2j}}{j(1-x^j)} \sim \frac{1}{1-x} \sum_{j=1}^{\infty} \frac{1}{j^2} \sim \frac{\pi^2}{6(1-x)}.$$

Therefore $\log p(n) \sim C\sqrt{n}$.

Hardy's proof of $p(n) \sim \frac{1}{4n\sqrt{3}} \exp(C\sqrt{n})$

Recall the definition of the Dedekind eta function:

$$\eta(\tau) = e^{\pi i \tau / 12} \prod_{n=1}^{\infty} (1 - e^{2\pi i n \tau}), \text{ for } \text{Im} \tau > 0.$$

Then $\eta(\tau) = z^{1/24} / f(z)$ where $z = e^{2\pi i \tau}$ and $\text{Im} \tau > 0$. The theory of modular forms tells us $\eta(-1/\tau) = \sqrt{\frac{i}{\tau}} \eta(\tau)$. Hence we obtain a functional equation for f :

$$f(z) = \frac{z^{1/24}}{\sqrt{2\pi}} \sqrt{\log(1/z)} \exp\left(\frac{\pi^2}{6 \log(1/z)}\right) f(z'), \text{ where } z' = \exp(-4\pi^2 / \log(1/z)). \quad (\star)$$

Note that $f(z') \rightarrow 1$ as $z \rightarrow 1$, so $(\star)$ basically gives a formula of $f(z)$ near $z = 1$.

Let $C_r = B(0, r)$ with $r \rightarrow 1^-$, then by Cauchy integral formula,

$$p(n) = \frac{1}{2\pi i} \int_C \frac{f(z)}{z^{n+1}} dz.$$

We parameterize C_r by $z = e^{-s+it}$ where $t \in [-\pi, \pi]$ and $e^{-s} = r$. Then

$$p(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(e^{-s+it}) e^{ns-int} dt$$

The major term when $t \rightarrow 0$ is $\exp\left(\frac{\pi^2}{6s} + ns\right)$, so we let $s = \frac{\pi}{\sqrt{6n}}$, then the term becomes $\exp(C\sqrt{n})$. Take $t_0 = n^{-7/10}$, and consider the major arc $|t| \leq t_0$, then for $\omega = s - it$,

$$I_1 = \frac{1}{2\pi} \int_{-t_0}^{t_0} f(e^{-s+it}) e^{ns-int} dt = \frac{1}{2\pi} \int_{-t_0}^{t_0} \frac{e^{-\omega/24}}{\sqrt{2\pi}} \sqrt{\omega} \exp\left(\frac{\pi^2}{6\omega} + n\omega\right) f(e^{-4\pi^2/\omega}) dt.$$

Since $\operatorname{Re}\left(\frac{1}{\omega}\right) = \frac{s}{s^2+t^2} \geq \frac{1}{2s} = c\sqrt{n}$, $z' = e^{-4\pi^2/\omega} \leq e^{-4\pi^2 c\sqrt{n}}$. Clearly $f(z) = 1 + O(z)$, so $f(e^{-4\pi^2/\omega}) = 1 + o\left(\frac{1}{n^2}\right)$. Additionally, $e^{-\omega/24} = 1 + O\left(\frac{1}{\sqrt{n}}\right)$, and $\sqrt{\omega} = \sqrt{s}(1 + o(1))$.

Note that

$$\begin{aligned} \frac{\pi^2}{6\omega} + n\omega &= \frac{\pi^2}{6s(1-it/s)} + n(s-it) = C\sqrt{n} + \frac{\pi^2}{6s} \left(\frac{it}{s} - \frac{t^2}{s^2} + O\left(\frac{t^3}{s^3}\right) \right) - int \\ &= C\sqrt{n} - \frac{n}{s} t^2 + O\left(\frac{t^3}{s^4}\right). \end{aligned}$$

By the choice of t_0 , $\frac{t_0^3}{s^4} \leq \frac{t_0^3}{s^4} = O(n^{-0.1}) = o(1)$. Therefore the integral becomes

$$I_1 = \frac{\sqrt{s} e^{C\sqrt{n}}}{2\pi\sqrt{2\pi}} \int_{-t_0}^{t_0} \exp\left(-\frac{n}{s} t^2\right) dt \sim \frac{\sqrt{s} e^{C\sqrt{n}}}{2\pi\sqrt{2\pi}} \int_{-\infty}^{\infty} \exp\left(-\frac{n}{s} t^2\right) dt = \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

(since $\int_{t_0}^{\infty} \exp\left(-\frac{n}{s} t^2\right) dt \leq \frac{s}{2nt_0} e^{-nt_0^2/s}$ is exponentially small because $\frac{nt_0^2}{s} \sim n^{0.1}$.)

Now we show the rest of the integral is small: Consider

$$\begin{aligned} \log f(e^{-s}) - \log|f(e^{-s+it})| &= \sum_{n=1}^{\infty} \frac{1}{n} \sum_{k=1}^{\infty} e^{-nks} (1 - \cos(nkt)) \geq \sum_{k=1}^{\infty} e^{-ks} (1 - \cos kt) \\ &= \operatorname{Re} \left(\frac{1}{e^s - 1} - \frac{1}{e^{s-it} - 1} \right) \sim \operatorname{Re} \left(\frac{1}{s-1} - \frac{1}{s-it-1} \right) \\ &= \frac{t^2}{s(s^2+t^2)} \geq cn^{0.1}. \end{aligned}$$

(Since $\frac{1}{e^z-1} - \frac{1}{z}$ is holomorphic, it is bounded.)

Hence $|f(e^{-s+it})| \leq \exp(-cn^{0.1}) f(e^{-s})$, so $|\int_{t_0}^{\pi} f(e^{-s+it}) e^{n(s-it)} dx| \leq e^{C\sqrt{n}} \pi f(e^{-s}) e^{-cn^{0.1}} = I_1 e^{-cn^{0.1}}$ is a lower term. Therefore

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right).$$

Ramanujan's proof of $p(n) = \sum_{q=1}^{\lfloor \alpha \sqrt{n} \rfloor} A_q(n) \phi(n) + O(n^{-1/4})$

Ramanujan considered the function $F(z) = \sum_{n=1}^{\infty} \Psi_a(n) z^n$, where for $a > 0$,

$$\Psi_a(n) = \frac{d}{dz} \frac{\cosh a \sqrt{z - 1/24} - 1}{\sqrt{z - 1/24}} \Big|_{z=n}$$

which is holomorphic for $\operatorname{Re} z \geq 1/24$. Recall Lindelöf's identity (see the proof of the prime number theorem), if $F(z)$ were holomorphic for $\operatorname{Re} z \geq \alpha$, then

$$F(z) = - \int_{\alpha-i\infty}^{\alpha+i\infty} \frac{\Psi_a(s) z^s}{e^{2\pi i s} - 1} ds.$$

Therefore let $\alpha = \frac{1}{24}$, we obtain

$$F(z) = z^{1/24} \int_{-\infty}^{\infty} \frac{z^{iy}}{1 - e^{i\pi/12 - 2\pi y}} \Psi_a(iy + 1/24) d(1/24 + iy) = I_1 + I_2 + I_3,$$

where if we let $aq = \pi \sqrt{2/3}$,

$$I_1(z) = iz^{1/24} \int_0^{\infty} z^{iy} \Psi_a(1/24 + iy) dy = \dots = z^{1/24} \sqrt{\pi} \sqrt{\log(1/z)} \left(e^{\pi^2/6q^2 \log(1/z)} - 1 \right),$$

$$I_2(z) = iz^{1/24} \int_0^{\infty} \frac{z^{iy}}{e^{2\pi y - i\pi/12} - 1} \Psi_a(1/24 + iy) dy,$$

$$I_3(z) = iz^{1/24} \int_0^{\infty} \frac{z^{-iy}}{1 - e^{2\pi y + i\pi/12}} \Psi_a(1/24 - iy) dy.$$

Note that I_2, I_3 is bounded as $|z| \rightarrow 1^-$.

To connect F with f , we again use the modular property of the Dedekind eta function.

Let $p, p', q \in \mathbb{N}$ with $p \leq q$, $\gcd(p, q) = 1$ and $q \mid 1 + pp'$, then for $s \in \mathbb{C}$, if $\tau = \frac{p}{q} + \frac{is}{q}$ and $T = \frac{p'}{q} + \frac{i}{qs}$, we have $T = \frac{c+d\tau}{a+b\tau}$ where $a = p, b = -q, c = \frac{1+pp'}{q}, d = -p'$ and $ad - bc = 1$. Hence we have

$$f(z) = \omega_{p,q} \sqrt{s} \exp \left(\frac{\pi}{12qs} - \frac{\pi s}{12q} \right) f(z')$$

where $z = e^{2\pi i \tau}$, $z' = e^{2\pi i T}$, and $\omega_{p,q}$ is a $24q$ -th root of unity.

Ramanujan noticed that if we let $z_{p,q} = \exp(-2\pi s/q)$ and $z'_{p,q} = \exp \left(-\frac{4\pi^2}{q^2 \log(1/z_{p,q})} + \frac{2\pi p' i}{q} \right)$, then $s = \frac{q}{2\pi} \log(1/z_{p,q})$ and $z_{p,q}^{1/24} = e^{-\pi s/12q}$. Hence

$$f(z) = \omega_{p,q} z_{p,q}^{1/24} \frac{\sqrt{q}}{\sqrt{2\pi}} \sqrt{\log(1/z_{p,q})} \exp \left(\frac{\pi^2}{6q^2 \log(1/z_{p,q})} \right) f(z'_{p,q}).$$

We denote the right-hand side as $\Omega(z_{p,q})$. Then as $z \rightarrow e^{2\pi i p/q}$, $z_{p,q} \rightarrow 1$ so $z'_{p,q} \rightarrow 0$ and $f(z'_{p,q}) \rightarrow 1$. Thus $\Omega(z_{p,q})$ behaves like $I_1(z)$, up to a factor of $\omega_{p,q}\sqrt{q}/\pi\sqrt{2}$.

For arbitrary $m \in \mathbb{N}$, summing up $1 \leq p \leq q \leq m$, $\gcd(p, q) = 1$ we obtain

$$\begin{aligned} f(z) - \sum_{q=1}^m \Omega(z_{p,q}) &\sim f(z) - \sum_{q=1}^m \omega_{p,q} \frac{\sqrt{q}}{\pi\sqrt{2}} F(z_{p,q}) \\ &\sim f(z) - \sum_{q=1}^m \omega_{p,q} \frac{\sqrt{q}}{\pi\sqrt{2}} \sum_{p=1}^q \Psi_{\pi\sqrt{2/3}/q}(n) (ze^{-2\pi i p/q})^n =: f(z) - \Phi(z). \end{aligned}$$

Define

$$A_q(n) = \sum_{p=1}^q \omega_{p,q} e^{-2\pi n p i/q}, \phi(n) = \frac{\sqrt{q}}{2\pi\sqrt{2}} \frac{de^{a\lambda_z/q}/\lambda_z}{dz} \Big|_{z=n}$$

where $a = \pi\sqrt{2/3}$ and $\lambda_z = \sqrt{z - 1/24}$.

then by Cauchy integral formula,

$$p(n) - \sum_{q=1}^m A_q(n) \phi(n) = \frac{1}{2\pi i} \int_{C_r} \frac{f(z) - \Phi(z)}{z^{n+1}} dz$$

By properties of the Farey sequence of order m , we can show that the integral has order $O(e^{\delta\sqrt{n}})$ for a constant δ .

Using calculations made by MacMahon, Ramanujan found the best approximation, by taking $m = \lfloor \alpha\sqrt{n} \rfloor$ we obtain

$$p(n) = \sum_{q=1}^{\lfloor \alpha\sqrt{n} \rfloor} A_q(n) \phi(n) + O(n^{-1/4}).$$

Proof from Stein

Stein gave another proof in his book:

$$p(n) = \frac{1}{2n\sqrt{2}} \frac{d}{dn} \left(\frac{e^{C\sqrt{n-1/24}}}{\sqrt{n-1/24}} \right) + O(e^{C\sqrt{n}/2})$$

where $C = \pi\sqrt{2/3}$.